Software Requirements Specification — Atlas Booking Service

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Introduction

The **Atlas Booking Service** is a fictional software system designed to streamline the process of booking travel services, including flights, hotels, and car rentals. This document outlines the functional and non-functional requirements for the system, ensuring alignment with stakeholder expectations and operational goals. The Software Requirements Specification (SRS) serves as a foundational document for development, testing, and validation, providing a clear understanding of the system's scope, capabilities, and constraints.

Purpose

The primary purpose of the Atlas Booking Service is to provide users with a centralized platform to search, compare, and book travel-related services efficiently. The system aims to enhance user experience by offering intuitive navigation, real-time availability checks, and secure transaction processing. Additionally, it supports administrative functions such as managing service providers, monitoring bookings, and generating reports.

Scope

The Atlas Booking Service will focus on the following core functionalities:

- User registration, authentication, and profile management.
- Search and comparison of travel services (flights, hotels, car rentals).
- Booking creation, modification, and cancellation.
- Payment processing and transaction tracking.
- Administrative tools for managing service providers and system settings.

The system will not include features such as real-time customer support, third-party service integration beyond payment gateways, or advanced analytics beyond basic reporting.

Definitions

- **User**: A registered individual or organization using the system to book travel services.
- **Service Provider**: A third-party entity (e.g., airline, hotel chain) offering travel services through the Atlas platform.
- **Booking**: A confirmed reservation of a travel service, including details such as dates, pricing, and availability.
- **Transaction**: A payment or refund process associated with a booking.

- **Admin**: A system administrator with elevated privileges to manage content, users, and service providers.

Functional Requirements

FR-1: User Registration and Authentication

Users must be able to register, log in, and authenticate their accounts using email and password. Multi-factor authentication (MFA) is optional but recommended for enhanced security.

FR-2: Service Search and Comparison

Users can search for travel services by location, date, price range, and service type. Results should include comparative metrics such as price, duration, and customer ratings.

FR-3: Booking Creation and Management

Users can create, edit, or cancel bookings. The system must validate availability, calculate prices, and confirm reservations via email.

FR-4: Payment Processing

Integrated payment gateways (e.g., Stripe, PayPal) will handle transactions. Users must enter payment details, and the system will confirm successful payments or refund cancellations.

FR-5: User Profile Management

Users can update personal information, view booking history, and manage payment methods. Profile data must be encrypted and accessible only to the user.

FR-6: Service Provider Management

Admins can add, edit, or remove service providers. The system must validate provider credentials and ensure service availability.

FR-7: Notification System

Users receive real-time notifications for booking confirmations, cancellations, and payment updates via email and in-app alerts.

FR-8: Reporting and Analytics

Admins can generate reports on bookings, revenue, and service provider performance. Data must be exportable in CSV or PDF formats.

Non-Functional Requirements

NFR-1: Performance

The system must respond to user requests within 2 seconds under normal load (1,000 concurrent users). Page load times should not exceed 3 seconds.

NFR-2: Security

All user data, including payment information, must be encrypted using AES-256. The system must comply with GDPR and PCI-DSS standards.

NFR-3: Scalability

The platform should support up to 10,000 concurrent users without performance degradation. Cloud-based infrastructure (e.g., AWS) is required for scalability.

NFR-4: Usability

The interface must be intuitive, with a mobile-responsive design. Users should complete a booking within 3 minutes.

NFR-5: Reliability

The system must maintain 99.9% uptime, with automatic failover mechanisms for critical components. Downtime must be minimized to less than 1 hour per month.

Assumptions

- Third-party payment gateways and service provider APIs will be available and stable.
- User data will be provided in a structured format (e.g., CSV, JSON) for initial setup.
- The system will operate in a regulated environment with compliance requirements for data privacy and financial transactions.

Constraints

- Development must adhere to a 12-month timeline with quarterly milestones.
- Budget limitations restrict the use of advanced AI-driven features.
- The system must be compatible with modern web browsers (Chrome, Firefox, Safari) and mobile operating systems (iOS, Android).

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